

TRAFFIC FLOW PREDICTION USING GRAPH NEURAL NETWORKS

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Year: 2024

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Degree Program: BSCS

Keywords: Graph Neural Networks, GNN, traffic prediction, Cavite, road network

ABSTRACT

This study applies a Graph Convolutional Network (GCN) on the Imus-Bacoor road network graph for traffic flow prediction, achieving a 15.2% improvement in RMSE over baseline LSTM models.

INTRODUCTION

Traffic congestion in Cavite's urban corridors causes significant economic and environmental costs. Traditional prediction models treat roads independently, ignoring the spatial relationships in road networks. Graph Neural Networks (GNNs) can model these dependencies explicitly. This study constructs a road network graph from OpenStreetMap data and applies GCN-based traffic flow prediction for the Imus-Bacoor corridor.

METHODOLOGY

The road network of the Imus-Bacoor corridor was modeled as a directed weighted graph (147 nodes, 298 edges) extracted from OpenStreetMap. Speed and volume data from 12 CCTV-based sensors were aggregated at 15-minute intervals over 6 months. A Graph Convolutional Network with 2 convolutional layers and a GRU temporal module was implemented in PyTorch Geometric. RMSE and MAE were computed against ground-truth sensor readings.

RESULTS

GCN+GRU achieved RMSE of 3.81 km/h vs. LSTM baseline of 4.49 km/h (15.2% improvement) and a plain GCN of 4.12 km/h. MAE improved from 3.21 to 2.74 km/h. The model captured bottleneck propagation effects that LSTM missed. Prediction accuracy degraded slightly during typhoon events (RMSE +0.61 km/h).

DISCUSSION

Capturing spatial road dependencies via GCN substantially improves traffic flow prediction. The improvement is most pronounced at network junction nodes where upstream flow strongly impacts downstream speed. Typhoon-period degradation suggests the model would benefit from weather covariate features. Future work should test real-time inference at 5-minute intervals.